Computing the mean of a random variable X can be a simple and effective way to establish the existence of a graph G such that $X(G) < a$ for some fixed $a > 0$ and, moreover, G has some desired property $\mathcal{P}$. Indeed, if the expected value of X is small, then $X(G)$ cannot be large for more than a few graphs in $\mathcal{G}(n,p)$, because $X(G) \geqslant 0$ for all $G \in \mathcal{G}(n,p)$. Hence X must be small for many graphs in $\mathcal{G}(n,p)$, and it is reasonable to expect that among these we may find one with the desired property $\mathcal{P}$.

This simple idea lies at the heart of countless non-constructive existence proofs using random graphs, including the proof of Erdős's theorem presented in the next section. Quantified, it takes the form of the following lemma, whose proof follows at once from the definition of the expectation and the additivity of $\mathbb{P}$:

[11.2.2]
[11.4.1]
[11.4.3]

Lemma 11.1.5. (Markov's Inequality)
Let $X \geqslant 0$ be a random variable on $\mathcal{G}(n,p)$ and $a > 0$. Then

$$\mathbb{P}\,[\,X \geqslant a\,] \leqslant \mathbb{E}(X)/a\,.$$

Proof.

$$\mathbb{E}(X) = \sum_{G \in \mathcal{G}(n,p)} \mathbb{P}(\{G\}) \cdot X(G) \;\geqslant \sum_{\substack{G \in \mathcal{G}(n,p) \\ X(G) \geqslant a}} \mathbb{P}(\{G\}) \cdot a = \mathbb{P}\,[\,X \geqslant a\,] \cdot a\,.$$

□

11.2 The probabilistic method

Very roughly, the *probabilistic method* in discrete mathematics has developed from the following idea. In order to prove the existence of an object with some desired property, one defines a probability space on some larger – and certainly non-empty – class of objects, and then shows that an element of this space has the desired property with positive probability. The 'objects' inhabiting this probability space may be of any kind: partitions or orderings of the vertices of some fixed graph arise as naturally as mappings, embeddings and, of course, graphs themselves. In this section, we illustrate the probabilistic method by giving a detailed account of one of its earliest results: of Erdős's classic theorem on large girth and chromatic number (Theorem 5.2.5).

Erdős's theorem says that, given any positive integer k, there is a graph G with girth $g(G) > k$ and chromatic number $\chi(G) > k$. Let us call cycles of length at most k *short*, and sets of $|G|/k$ or more vertices *short*
big. For a proof of Erdős's theorem, it suffices to find a graph G without *big/small*
short cycles and without big independent sets of vertices: then the colour classes in any vertex colouring of G are *small* (not big), so we need more than k colours to colour G.

How can we find such a graph G? If we choose p small enough, then a random graph in $\mathcal{G}(n,p)$ is unlikely to contain any (short) cycles. If we choose p large enough, then G is unlikely to have big independent vertex sets. So the question is: do these two ranges of p overlap, that is, can we choose p so that, for some n, it is both small enough to give $\mathbb{P}[\, g \leqslant k \,] < \frac{1}{2}$ and large enough for $\mathbb{P}[\, \alpha \geqslant n/k \,] < \frac{1}{2}$? If so, then $\mathcal{G}(n,p)$ will contain at least one graph without either short cycles or big independent sets.

Unfortunately, such a choice of p is impossible: the two ranges of p do not overlap! As we shall see in Section 11.4, we must keep p below n^{-1} to make the occurrence of short cycles in G unlikely – but for any such p there will most likely be no cycles in G at all (Exercise 18), so G will be bipartite and hence have at least $n/2$ independent vertices.

But all is not lost. In order to make big independent sets unlikely, we shall fix p above n^{-1}, at $n^{\epsilon-1}$ for some $\epsilon > 0$. Fortunately, though, if ϵ is small enough then this will produce only *few* short cycles in G, even compared with n (rather than, more typically, with n^k). If we then delete a vertex in each of those cycles, the graph H obtained will have no short cycles, and its independence number $\alpha(H)$ will be at most that of G. Since H is not much smaller than G, its chromatic number will thus still be large, so we have found a graph with both large girth and large chromatic number.

To prepare for the formal proof of Erdős's theorem, we first show that an edge probability of $p = n^{\epsilon-1}$ is indeed always large enough to ensure that $G \in \mathcal{G}(n,p)$ 'almost surely' has no big independent set of vertices. More precisely, we prove the following stronger assertion:

Lemma 11.2.1. *Let $k > 0$ be an integer, and let $p = p(n)$ be a function of n such that $p(n) \geqslant 16k^2/n$ for n large. Then*

$$\lim_{n\to\infty} \mathbb{P}[\, \alpha \geqslant \tfrac{1}{2}n/k \,] = 0\,.$$

(11.1.2) *Proof.* For all integers $n \geqslant r \geqslant 2$ and $G \in \mathcal{G}(n,p)$, Lemma 11.1.2 implies[3]

$$\mathbb{P}[\, \alpha \geqslant r \,] \leqslant \binom{n}{r} q^{\binom{r}{2}} \leqslant 2^n q^{\binom{r}{2}} \leqslant 2^n e^{-p\binom{r}{2}}.$$

Hence if $p \geqslant 16k^2/n$ and $r \geqslant \frac{1}{2}n/k \geqslant 2$, then

$$\mathbb{P}[\, \alpha \geqslant r \,] \leqslant 2^n e^{-pr^2/4} \leqslant 2^n e^{-pn^2/16k^2} \leqslant 2^n e^{-n} \underset{n\to\infty}{\longrightarrow} 0\,.$$

As α is an integer and thus $\mathbb{P}[\, \alpha \geqslant r \,] = \mathbb{P}[\, \alpha \geqslant \frac{1}{2}n/k \,]$ for $r = \lceil \frac{1}{2}n/k \rceil$, this implies the assertion. □

[3] To see the second inequality, count the subsets of an n-set. For the third, note that $1 - p \leqslant e^{-p}$ for all p: compare the functions $x \mapsto e^x$ and $x \mapsto x + 1$ for $x = -p$.

We are now ready to prove Theorem 5.2.5, which we restate:

Theorem 11.2.2. (Erdős 1959) [9.2.3]
For every integer k there exists a graph H with girth $g(H) > k$ and chromatic number $\chi(H) > k$.

(11.1.5)
Proof. Assume that $k \geqslant 3$, fix ϵ with $0 < \epsilon < 1/k$, and let $p := n^{\epsilon - 1}$. Let (11.1.4)
$X(G)$ denote the number of short cycles in a random graph $G \in \mathcal{G}(n,p)$, p, ϵ, X
i.e. its number of cycles of length at most k.

By Lemma 11.1.4, we have

$$\mathbb{E}(X) = \sum_{i=3}^{k} \frac{(n)_i}{2i} p^i \leqslant \tfrac{1}{2} \sum_{i=3}^{k} n^i p^i \leqslant \tfrac{1}{2}(k-2)\, n^k p^k\,;$$

note that $(np)^i \leqslant (np)^k$, because $np = n^\epsilon \geqslant 1$. By Lemma 11.1.5,

$$\begin{aligned} \mathbb{P}\,[\, X \geqslant n/2 \,] &\leqslant \mathbb{E}(X)\big/(n/2) \\ &\leqslant (k-2)\, n^{k-1} p^k \\ &= (k-2)\, n^{k-1} n^{(\epsilon-1)k} \\ &= (k-2)\, n^{k\epsilon - 1}. \end{aligned}$$

As $k\epsilon - 1 < 0$ by our choice of ϵ, this implies that

$$\lim_{n\to\infty} \mathbb{P}\,[\, X \geqslant n/2 \,] = 0\,.$$

Let n be large enough that $\mathbb{P}\,[\, X \geqslant n/2 \,] < \frac{1}{2}$ and $\mathbb{P}\,[\, \alpha \geqslant \frac{1}{2} n/k \,] < \frac{1}{2}$; n
the latter is possible by our choice of p and Lemma 11.2.1. Then there is a graph $G \in \mathcal{G}(n,p)$ with fewer than $n/2$ short cycles and $\alpha(G) < \frac{1}{2} n/k$. From each of those cycles delete a vertex, and let H be the graph obtained. Then $|H| \geqslant n/2$ and H has no short cycles, so $g(H) > k$. By definition of G,

$$\chi(H) \geqslant \frac{|H|}{\alpha(H)} \geqslant \frac{n/2}{\alpha(G)} > k\,.$$

□

Corollary 11.2.3. *There are graphs with arbitrarily large girth and arbitrarily large values of the invariants κ, ε and δ.*

Proof. Apply Lemma 5.2.3 and Theorem 1.4.3. □ (1.4.3)
(5.2.3)

11.3 Properties of almost all graphs

Recall that a *graph property* is a class of graphs that is closed under isomorphism, one that contains with every graph G also the graphs isomorphic to G. If $p = p(n)$ is a fixed function (possibly constant), and $\mathcal{P}$ is a graph property, we may ask how the probability $\mathbb{P}[\, G \in \mathcal{P}\,]$ behaves for $G \in \mathcal{G}(n,p)$ as $n \to \infty$. If this probability tends to 1, we say that $G \in \mathcal{P}$ for *almost all* (or *almost every*) $G \in \mathcal{G}(n,p)$, or that $G \in \mathcal{P}$ *almost surely*; if it tends to 0, we say that *almost no* $G \in \mathcal{G}(n,p)$ has the property $\mathcal{P}$. (For example, in Lemma 11.2.1 we proved that, for a certain p, almost no $G \in \mathcal{G}(n,p)$ has a set of more than $\frac{1}{2}n/k$ independent vertices.)

almost all etc.

To illustrate the new concept let us show that, for constant p, every fixed abstract[4] graph H is an induced subgraph of almost all graphs:

Proposition 11.3.1. *For every constant $p \in (0,1)$ and every graph H, almost every $G \in \mathcal{G}(n,p)$ contains an induced copy of H.*

Proof. Let H be given, and $k := |H|$. If $n \geqslant k$ and $U \subseteq \{0,\dots,n-1\}$ is a fixed set of k vertices of G, then $G[U]$ is isomorphic to H with a certain probability $r > 0$. This probability r depends on p, but not on n (why not?). Now G contains a collection of $\lfloor n/k \rfloor$ disjoint such sets U. The probability that none of the corresponding graphs $G[U]$ is isomorphic to H is $(1-r)^{\lfloor n/k \rfloor}$, since these events are independent by the disjointness of the edges sets $[U]^2$. Thus

$$\mathbb{P}[\, H \nsubseteq G \text{ induced}\,] \leqslant (1-r)^{\lfloor n/k \rfloor} \underset{n\to\infty}{\longrightarrow} 0\,,$$

which implies the assertion. □

The following lemma is a simple device enabling us to deduce that quite a number of natural graph properties (including that of Proposition 11.3.1) are shared by almost all graphs. Given $i,j \in \mathbb{N}$, let $\mathcal{P}_{i,j}$ denote the property that the graph considered contains, for any disjoint vertex sets U, W with $|U| \leqslant i$ and $|W| \leqslant j$, a vertex $v \notin U \cup W$ that is adjacent to all the vertices in U but to none in W.

$\mathcal{P}_{i,j}$

Lemma 11.3.2. *For every constant $p \in (0,1)$ and $i,j \in \mathbb{N}$, almost every graph $G \in \mathcal{G}(n,p)$ has the property $\mathcal{P}_{i,j}$.*

[4] The word 'abstract' is used to indicate that only the isomorphism type of H is known or relevant, not its actual vertex and edge sets. In our context, it indicates that the word 'subgraph' is used in the usual sense of 'isomorphic to a subgraph'.